Multi-Resolution Transformer-Augmented Efficientnet for Explainable Breast Cancer Histopathology Analysis

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Abstract:
Diagnosis of breast cancer based on the histopathology images requires clinical reliability that is based on accurate and interpretable models. The proposed study is Multi-Resolution Transformer-Augmented EfficientNet (MRTAE) framework, a framework that combines EfficientNet to extract scale-wise features with a transformer layer to resolve long-range spatial relationships across scales. The model improves contextual perception and offers explainable predictions by use of Grad-CAM++ visualization. Architecture tests on benchmark datasets like BreakHis and Camelyon17 show excellent performance in accuracy, F1-score and interpretability over the traditional CNN-based ones. The suggested MR-TAE can be considered a good compromise between diagnostic accuracy and transparency as a high-tech tool that can help pathologists in the detection and classification of breast cancer.

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